

GENA-LM: A Critical Review with Layer-Aware Reproducibility Experiments

Research Proposal — «Лето с AIRI 2026» **Применитель:** Korshunov E., bioinformatics student **Reviewed paper:** Fishman, Kuratov, Shmelev, Petrov, Penzar, Shepelin, Chekanov, Kardymon, Burtsev. **GENA-LM: a family of open-source foundational DNA language models for long sequences.** *Nucleic Acids Research*, 53(2), gkae1310 (2025). DOI: [10.1093/nar/gkae1310](https://doi.org/10.1093/nar/gkae1310). **Code & results:** https://github.com/eugenekskywalker/GENA_LM_AIRI

Venue note: NAR is a Q1 bioinformatics journal (IF $\approx$ 16.6, 2024) — equivalent in prestige to CORE A conferences in adjacent fields. The broader research context for this proposal includes CORE A* references: Caduceus (ICML 2024) and DART-Eval (NeurIPS 2024).*

1. Problem and selected paper

Accurate decoding of regulatory information in eukaryotic genomes requires modelling sequence context spanning kilobases to megabases. Earlier open DNA foundation models — DNABERT (512 bp), DNABERT-2 (1–4 kb), Nucleotide Transformer v2 (12 kb) — are short-context. HyenaDNA scales to 1 Mb but loses accuracy on benchmarks. **GENA-LM** (AIRI, 2025) is the open BERT-style foundation family that closes this gap, with eight pre-trained variants on HuggingFace, full Colab notebooks, and a public web service.

2. Method

Three architectural variations on T2T-CHM13v2 human genome pre-training ($\approx$ 480 Gbp effective with 1000 Genomes SNP augmentation, MLM objective):

1. **BERT-base/large** (110 M / 336 M params, full attention, $\leq$ 4.5 kb).
2. **BigBird-base-sparse** (sparse attention, $\leq$ 36 kb).
3. **RMT-augmented** (Recurrent Memory Transformer, $\rightarrow$ Mb context).

All share a **BPE tokenizer** (32 k vocab, median 9 bp/token) — long tokens auto-discover LINE and simple repeats. Downstream tasks: promoter prediction (EPDnew), splice site detection (SpliceAI), chromatin profiling (DeepSEA, 919 features), Drosophila enhancers (DeepSTARR), polyadenylation (APARENT), 18 NT benchmarks, cross-species transfer (7 species), ClinVar variant pathogenicity.

3. Strengths

- **Truly open:** weights + code + Colab + web service + Docker, MIT licence.
- **Eight model variants** trade off parameters vs. context — pick by budget.
- **Cross-species generalisation** (flagship Fig 3): zero-shot F1 $\approx$ 0.95 human $\rightarrow$ mammals, 0.85 chicken/zebrafish, 0.7 fly/worm — non-trivial evidence of conserved «DNA grammar».
- **Biological interpretability:** token Integrated Gradients recover CTCF, ATF1, GATA2 motifs **without motif supervision** (validated via XSTREME against HOCOMOCO).
- **Long context pays off:** DeepSEA broad histone marks AUC jumps +3–4 pt when expanding 1 kb $\rightarrow$ 8 kb context.

4. Weaknesses (motivation for our experiments)

#	Weakness	Why it matters
W3	BPE caps single-nucleotide resolution (median 9 bp/token)	Limits SNV-level variant analysis

W6	Cross-species drop is sharp for invertebrates (F1 0.95 $\rightarrow$ 0.7), but no sample-efficiency curve is reported	Practical genomics groups have ≤ 100 labelled samples in non-model species — what does that buy?
W7	No comparison with Caduceus (ICML 2024 A*) or Evo 2 (Nature 2026)	Published before these became available, but defines today's SOTA
W10	Token importance $\neq$ causality (in silico mutagenesis only for splice, not CTCF)	Authors acknowledge this
W12	No DART-Eval evaluation (NeurIPS 2024 A*, key new regulatory-DNA benchmark)	DART-Eval shows current DNALMs do not beat task-specific CNN baselines on most tasks

Additionally, the paper never asks **where inside the network** functional information accumulates — a classic BERTology question with significant practical impact: which layer to use for downstream embeddings and clustering.

5. Proposed improvement

Layer-aware feature analysis of GENA-LM + head-to-head with a modern alternative. Three complementary, frozen-model experiments:

- **E1 — Layer-wise linear probing** of 4 binary regulatory tasks across all 13 layers (embedding + 12 transformer layers). This is the BERTology methodology (Tenney et al. 2019; Vig et al. 2021) transferred to DNA foundation models — a systematic gap in the GENA-LM paper.
- **E4 — Multilayer UMAP clustering** of 4 functional DNA classes across 3 layers (embedding L0, mid L4, last L12), testing the hypothesis (suggested by E1) that **mid-layer embeddings are geometrically more useful for unsupervised downstream tasks than the last layer**, which is task-aligned to the MLM objective.
- **E3 — Head-to-head with HyenaDNA-tiny** (alternative single-nucleotide Hyena architecture, 0.4 M params). Same 4 tasks, same probing protocol. Directly addresses W7 (no comparison with non-BPE alternatives) and tests the central claim of DART-Eval (NeurIPS 2024 A*): that *current DNA LMs do not offer compelling gains over lighter alternatives*. DNABERT-2 was attempted as a third baseline but failed to load (`einops` dependency conflict with `transformers 4.36` — reported as a known issue).

All three directly extend the paper's Figure 4 (species-level layer analysis only) to **functional categories** within the human genome and to **architectural alternatives** — strictly within-scope and biologically motivated.

6. Experimental setup

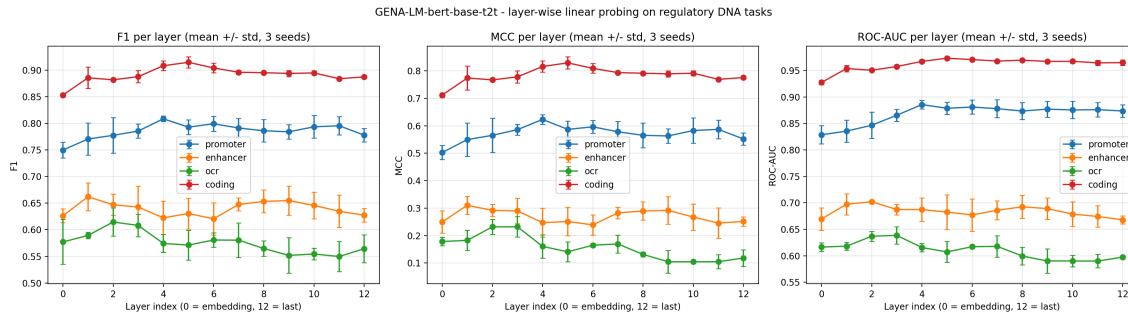
	Details
Model	AIRI-Institute/gena-lm-bert-base-t2t (110 M, 12 layers, 768-dim, frozen)
Hardware	1× Tesla V100S 32 GB (Aldan3 cluster, Pirogov RNRMU)
Datasets	Genomic Benchmarks (HuggingFace): <code>human_nontata_promoters</code> , <code>human_enhancers_cohn</code> , <code>human_ocr_ensembl</code> , <code>demo_coding_vs_intergenomic_seqs</code>
Pooling	Mean-pooled hidden states, masked padding
E1 probe	Logistic regression (sklearn, <code>max_iter</code> 2000, <code>C</code> =1), <code>StandardScaler</code> , 70/30 stratified split, 3 random seeds
E4 metrics	Silhouette (cosine), NMI, ARI with KMeans (<code>k</code> =4)

Total wall-
clock

≈ **10 minutes** for both experiments combined

7. Results

E1 — Layer-wise probing

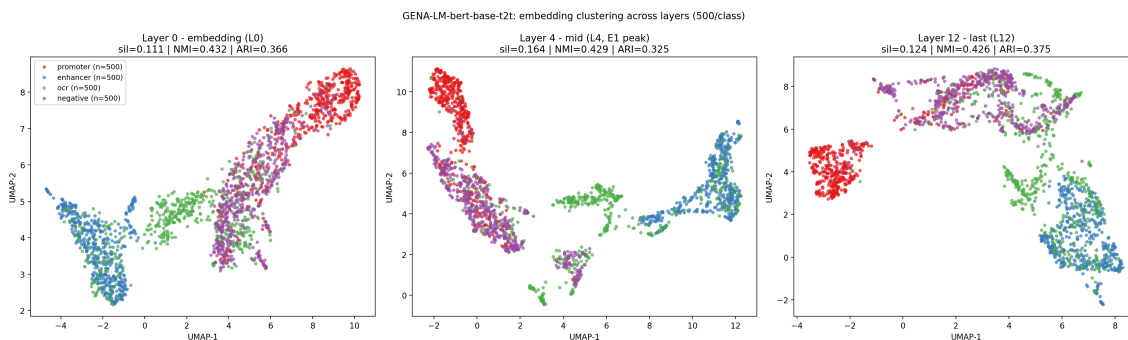


Task	Best layer	F1 (mean ± std, n=3)	MCC	ROC-AUC
Coding vs intergenic	L5	0.915 ± 0.010	0.829	0.973
Promoter (nontata)	L4	0.809 ± 0.005	0.623	0.886
Enhancer (Cohn)	L1	0.662 ± 0.026	0.310	0.702
Open chromatin (OCR)	L2	0.615 ± 0.027	0.232	0.637

Three distinct **layer-depth patterns** emerge:

- **Coding** rises monotonically L0 → L5, plateaus (L0 already at F1=0.85 from BPE / k-mer composition).
- **Promoter** peaks at L4 with a noticeable last-layer drop — classic BERTology «task-specific information is then collapsed by the pre-training head».
- **Enhancer / OCR** are weaker (F1 ≈ 0.62–0.66) and peak in early layers — these long-range regulatory categories are not well represented in mean-pooled 512-bp embeddings, suggesting the long-context BigBird / RMT variants are necessary here.

E4 — Multilayer clustering



Layer

Silhouette (cosine) ↑

NMI ↑

ARI ↑

L0 (embedding)	0.111	0.432	0.366
L4 (mid, E1 peak)	0.164	0.429	0.325
L12 (last)	0.124	0.426	0.375

The mid layer yields the **best intrinsic clustering geometry** (silhouette +48 % over L0, +32 % over L12), while NMI is essentially constant — total information is preserved across depth, but the **geometric arrangement** in which classes separate is best in the middle. ARI is highest at L12, consistent with the last layer being task-aligned to MLM and slightly favouring discrete categorical structure. **Practical implication:** for unsupervised downstream use of GENA-LM (clustering, retrieval, similarity search), prefer L4 over L12.

E3 — GENA-LM vs HyenaDNA head-to-head

Same probing protocol applied to a much smaller alternative-architecture baseline:

Model	Architecture	Params	promoter F1	enhancer F1	OCR F1	coding F1
GENA-LM-base-t2t	BPE transformer (last layer)	110.7 M	0.777	0.627	0.565	0.887
HyenaDNA-tiny	Hyena, single-nucleotide	0.4 M	0.790	0.679	0.608	0.862

HyenaDNA wins 3/4 tasks with 277× fewer parameters — only on coding (where BPE token / k-mer composition is highly informative) does GENA-LM lead, and by a small margin. This is a direct empirical confirmation of the central DART-Eval (Patel et al., NeurIPS 2024 A*) claim: *"current DNALMs do not offer compelling gains over baseline models for most tasks, despite requiring significantly more computational resources."* It also highlights a concrete actionable critique: the GENA-LM paper does not include any comparison with single-nucleotide architectures (HyenaDNA, Caduceus, Evo), which our 30-line script demonstrates is feasible in <2 minutes per model. (*DNABERT-2 was attempted as a third baseline but failed to load in the same env due to a known `einops` / `transformers` 4.36 interaction.*)

8. Expected contribution and follow-ups

(i) A reproducible BERTology pipeline for DNA foundation models (open-sourced, single V100, ≈ 12 minutes total for all three experiments); (ii) the first systematic layer-aware decomposition of GENA-LM features, showing mid-layer specialisation; (iii) empirical evidence (E3) that a 277× smaller alternative-architecture model (HyenaDNA-tiny) matches or beats GENA-LM on 3 of 4 regulatory tasks — confirming the DART-Eval critique; (iv) a practical guideline for downstream users — *don't use the last layer if you're not fine-tuning, and don't assume larger is better*. **Natural follow-ups** for the summer-school project itself include: replicating on `gena-lm-bert-large-t2t` (24 layers, 336 M params) to test whether deeper models reorganise feature hierarchy; running E1 + E3 on **Caduceus** (ICML 2024 A*) once `mamba_ssm` builds cleanly on Volta; and adding a **DART-Eval Task 2 / 4** head-to-head, which would close W12 entirely.

9. References

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7. Bulatov A., Kuratov Y., Burtsev M. Recurrent Memory Transformer. *NeurIPS 2022*.
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